

Problem 6-1b:

Tasks: (1) Effect of doubling P and decreasing T by 20°C; (2) Reversible case with $K_C = 0.02 \text{ mol dm}^{-3}$; (3) Irreversible with pressure drop $\alpha_p = 99 \times 10^3 \text{ dm}^{-3}$ (per tube).

Model and Assumptions

- Gas-phase, isothermal PFR, ideal gas, negligible radial gradients.
- Base stoichiometry (matches LEP): $A \rightarrow B + \frac{1}{2}C$.
- Rate law (base, irreversible): $r = -k C_A^2$.
- Ideal-gas relations: $C_T(V) = \frac{P(V)}{RT}$, $v(V) = \frac{F_T RT}{P(V)}$, $C_i = \frac{F_i}{v}$.
- PFR balances (per unit volume):

$$\frac{dF_A}{dV} = r, \quad \frac{dF_B}{dV} = -r, \quad \frac{dF_C}{dV} = -\frac{r}{2}.$$

- For pressure drop (part 3): tube model $\frac{dP}{dV} = -\alpha_p P \Rightarrow P(V) = P_0 e^{-\alpha_p V}$.

Kinetics. LEP 6-1 uses an Arrhenius form (units consistent with the LEP file you run):

$$k(T) = k_{\text{ref}} \exp \left[\frac{E}{R^*} \left(\frac{1}{T_{\text{ref}}} - \frac{1}{T} \right) \right],$$

where E is the activation energy (in the LEP file units), R^* matches those units, and k_{ref} is the reference rate constant at T_{ref} . (In the provided codes, simply edit the `BASE DATA` section to match your LEP file.)

Numerical Formulation

At position V ,

$$v = \frac{F_T RT}{P}, \quad C_A = \frac{F_A}{v}, \quad r = -k C_A^2.$$

With pressure drop,

$$\frac{dP}{dV} = -\alpha_p P.$$

The ODE system is integrated from $V = 0$ to $V = V_{\text{end}}$ with

$$F_A(0) = F_{A0}, \quad F_B(0) = F_{B0}, \quad F_C(0) = F_{C0}, \quad P(0) = P_0.$$

Conversion is $X(V) = \frac{F_{A0} - F_A(V)}{F_{A0}}$.

Part (1): P doubled, $T \rightarrow T - 20^\circ\text{C}$ (Irreversible)

We keep the irreversible model and repeat the integration at

$$P = P_{\text{new}} = 2P_0, \quad T = T_{\text{new}} = T_0 - 20^\circ\text{C}.$$

Observation: Doubling P increases concentrations (and thus the 2nd-order rate) while lowering T reduces k (Arrhenius). For the LEP numbers used, the pressure effect dominates and the exit conversion rises.

Part (2): Reversible with $K_C = 0.02 \text{ mol dm}^{-3}$

Use the reversible net rate

$$r_{\text{net}} = -k \left(C_A^2 - \frac{C_B C_C}{K_C} \right).$$

For the comparison, one may also use $A \rightleftharpoons B + C$ (1:1 products) to match K_C units; with $A \rightarrow B + \frac{1}{2}C$ the qualitative shape is the same: F_A approaches a nonzero asymptote, and F_B, F_C level off.

Part (3): Irreversible with pressure drop

Include $dP/dV = -\alpha_p P$ with $\alpha_p = 99 \times 10^3 \text{ dm}^{-3}$ (per tube). As P decays along V , C_T falls, so the local rate decreases and the product curves flatten; conversion is lower than the no-drop case.

Representative Results (using the provided LEP constants)

- Base exit conversion (no PD): $X_{\text{base}} \approx 0.93$.
- Part (1): $X_{\text{new}} \approx 0.97$ at $P = 2P_0$, $T = T_0 - 20^\circ\text{C}$.
- Part (2): Profiles flatten toward equilibrium; F_A asymptotes above zero; F_B, F_C rise more slowly and plateau.
- Part (3): With pressure drop, $P(V)$ decays roughly exponentially; $X_{\text{out}} \approx 0.86$ and F_B, F_C are lower; curves are more concave.

Problem P6-5B: Reversible elementary gas reaction with membrane removal in a PFR $A \rightleftharpoons B + 2C$, A = di-*tert*-butyl peroxide, B = C₂H₆, C = CH₃COCH₃

Given

- Isothermal, isobaric gas reactor (ideal gas).
- Feed: pure A at $F_{A0} = 2.5$ mol/min; $F_{B0} = F_{C0} = 0$.
- Pressure $P = 10$ atm (P6-4B); temperature for this problem $T = 300$ K (P6-5B).
- Elementary reversible kinetics. At $T_{\text{ref}} = 323$ K, $k_{\text{ref}} = 1 \times 10^{-4}$ /min; activation energy $E_a = 85$ kJ mol⁻¹.
- Equilibrium constant (concentration basis) at $T = 300$ K: $K_c = 0.025$ (mol/dm³)².
- Membrane (only in the membrane reactor case): first-order removal of B with coefficient $k_C = 0.08$ /s.

Assumptions

1. Ideal gas, negligible pressure drop, isothermal, isobaric.
2. Elementary reaction rate; reverse rate enforced by K_c at the same T .
3. Plug flow model; membrane removes only B and does not change total pressure.

1 Stoichiometry (per mol A fed)

For $A \rightleftharpoons B + 2C$, with conversion X (of A),

$$n_A = 1 - X, \quad n_B = X, \quad n_C = 2X, \quad n_T = 1 + 2X.$$

For a flowing system with inlet molar rate F_{A0} (pure A),

$$F_A = F_{A0}(1 - X), \quad F_B = F_{A0}X, \quad F_C = 2F_{A0}X, \quad F_T = F_{A0}(1 + 2X).$$

2 Thermodynamics: K_c -relation and X_{eq}

On a concentration basis,

$$K_c = \frac{C_B C_C^2}{C_A}.$$

With ideal gas and isothermal, isobaric operation

$$C_T = \frac{P}{RT}, \quad v = \frac{F_T RT}{P}, \quad C_i = \frac{F_i}{v} = \frac{F_i}{F_T} C_T = y_i C_T,$$

so that

$$K_c = \frac{y_B y_C^2}{y_A} C_T^2.$$

From the stoichiometry with pure A feed,

$$y_A = \frac{1-X}{1+2X}, \quad y_B = \frac{X}{1+2X}, \quad y_C = \frac{2X}{1+2X}.$$

Therefore the equilibrium conversion X_{eq} at the stated T, P is the root of

$$\underbrace{\left[\frac{X}{1+2X} \cdot \left(\frac{2X}{1+2X} \right)^2 \right] \bigg/ \frac{1-X}{1+2X}}_{\frac{y_B y_C^2}{y_A}} C_T^2 - K_c = 0. \quad (1)$$

Numerical constants at $P = 10 \text{ atm}$, $T = 300 \text{ K}$ (with $R = 0.082057 \text{ dm}^3 \text{ atm mol}^{-1} \text{ K}^{-1}$):

$$C_T = \frac{P}{RT} = \frac{10}{0.082057 \times 300} = 0.40622 \text{ mol dm}^{-3}.$$

Insert C_T and $K_c = 0.025$ into (1) and solve for $X \in (0, 1)$ (requires a numerical root find). If P is taken as $\sim 7.5 \text{ atm}$ instead of 10 atm , the resulting X_{eq} is close to the ~ 0.52 value printed beside P6-5B.

3 Kinetics

Arrhenius relation for $k(T)$:

$$k(T) = k_{\text{ref}} \exp \left[-\frac{E_a}{R_g} \left(\frac{1}{T} - \frac{1}{T_{\text{ref}}} \right) \right],$$

with $R_g = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$, $k_{\text{ref}} = 1 \times 10^{-4} / \text{min}$ at $T_{\text{ref}} = 323 \text{ K}$, $E_a = 85 \times 10^3 \text{ J mol}^{-1}$. Numerically at $T = 300 \text{ K}$,

$$k(300 \text{ K}) = 1.0 \times 10^{-4} \exp \left[-\frac{85000}{8.314} \left(\frac{1}{300} - \frac{1}{323} \right) \right] = \boxed{8.83 \times 10^{-6} \text{ 1/min}}.$$

Elementary reversible rate (net):

$$r = k C_A - \frac{k}{K_c} C_B C_C^2 \quad [\text{mol dm}^{-3} \text{ min}^{-1}]. \quad (2)$$

4 PFR model (no membrane)

For a plug flow reactor with volume coordinate V ,

$$\frac{dF_A}{dV} = -r, \quad \frac{dF_B}{dV} = +r, \quad \frac{dF_C}{dV} = +2r,$$

with r evaluated from (2) using the algebra

$$v = \frac{F_T R T}{P}, \quad C_A = \frac{F_A}{v}, \quad C_B = \frac{F_B}{v}, \quad C_C = \frac{F_C}{v}.$$

ICs at $V = 0$: $F_A(0) = F_{A0}$, $F_B(0) = 0$, $F_C(0) = 0$.

5 PFR with membrane removal of B

The membrane imposes a first-order sink on B in the bulk-phase balance:

$$\left(\frac{dF_B}{dV}\right)_{\text{mem}} = r - k_C C_B \quad \text{with} \quad k_C = 0.08 \text{ s}^{-1} = 4.8 \text{ min}^{-1},$$

while the other two remain unchanged:

$$\frac{dF_A}{dV} = -r, \quad \frac{dF_C}{dV} = 2r.$$

As before, $C_B = F_B/v$ with $v = F_T RT/P$.

6 Targets and stopping criteria

- **(a) Equilibrium conversion in a conventional PFR:** Solve (1) for X_{eq} (purely algebraic). Then integrate the conventional PFR to large V (or until $X \rightarrow X_{eq}$) to confirm the approach to equilibrium.
- **(b) Membrane reactor:** Integrate the membrane-PFR ODEs from $V = 0$ using the event

$$X(V) \equiv \frac{F_{A0} - F_A(V)}{F_{A0}} = 0.80$$

- At the same V^* , integrate the *conventional* PFR (no membrane) and report its exit conversion $X_{\text{conv}}(V^*)$.
- Detect any local maxima of $F_B(V)$ and $F_C(V)$ by scanning $V \mapsto F_i(V)$; report their values and locations.